MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCFSQ100 - FOOD AND NUTRITION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 50

Part A

MCQ

Answer all questionsEach question carries 1 marks

1. Identify the purpose of bomb calorimeter. [K] / [CO1, CO2] , (0.25mks , 0.75mks)

c). To measure energy content of sample

2.Identify incomplete protein from the following [U] / [CO1] (1 mark)

d. Wheat

3. B-complex vitamins important for athletes because[U] / [CO1, CO4](0.5mks, 0.5mks)

b. They aid in energy production and muscle function

4. Select which of the following is a macronutrient? [K] / [CO1] (1 mark)

c. Carbohydrate

5. Identify importance to follow the guidelines of the Food Guide Pyramid[K] / [CO1](1 mark)

b.To achieve a balanced and nutritious diet.

6. The main components of starch are[U] / [CO1](1 mark)

a.Amylose and Amylopectin

7. Identify the primary dietary source of carbohydrates[K] / [CO1](1 mark)

c. Cereals and grains

8. Which group of people is recommended to take calcium supplements?[U] / [CO1, CO3,CO4] (0.33mks,0.33mks ,0.33mks)

d). All of the above

9. High dietary cholesterol is caused due to[K] / [CO1](1 mark)

a. Intake of saturated fatty acid/c. trans fatty acids

10.What is the recommended daily intake of iron for adult women? [U] / [CO1, CO3,CO4]

c). 18 mg**(0.33mks, 0.33.mks,0.33mks)**

[1x10 = 10]

Part B

Short essay

Answer any 4 questions .Each question carries 5 marks

11. Explain malnutrition [K] / [CO1]

(5mks)

Definition of malnutrition

Classification

Symptoms and causative factors

Viscious cycle to be drawn for causative factors

Malnutrition refers to a condition resulting from an imbalance in nutrient intake, either insufficient or excessive. This can be caused by a lack of essential nutrients (undernutrition), or by consuming too many nutrients, potentially leading to obesity or related health issues (overnutrition). Malnutrition can manifest in various forms, including wasting, stunting, and mental changes.

The most common types of malnutrition are Kwashiorkor , Marasmus and Marasmic Kwashiorkor. Kwashiorkor is a severe form of malnutrition, primarily affecting children, caused by a lack of protein in the diet, despite sufficient calorie intake. It's characterized by edema (swelling), particularly in the abdomen and extremities, along with other symptoms like skin changes, hair discoloration, and reduced muscle mass. Cause:

Kwashiorkor arises from a deficiency in protein, even when the individual has enough carbohydrates and fats for energy.

Appearance:

Affected individuals, especially children, may appear emaciated with a protruding abdomen due to fluid retention (edema).

Symptoms:

Edema: Swelling of the feet, ankles, and abdomen is a key indicator. **Skin and Hair Changes**: Skin may become dry, scaly, and pigmented, and hair may lose its color and become brittle. **Loss of Muscle Mass**: Underweight individuals can have visible loss of muscle mass. **Other Symptoms**: Irritability, lethargy, and a weakened immune system are also common.

Consequences:

Kwashiorkor can lead to severe health complications, including increased susceptibility to infections, growth retardation, and in severe cases, death, if not treated promptly.

Prevalence:

It is most common in developing countries where access to protein-rich foods is limited, especially in areas where the diet relies heavily on carbohydrates like rice or maize

Marasmus is a severe form of malnutrition caused by a lack of calories and nutrients, leading to extreme emaciation and a weakened body. It's characterized by a severe wasting of muscle and fat tissue, stunting of growth, and a weakened immune system. The condition is often seen in young children, especially those who are not properly fed or have diseases that interfere with nutrient absorption.

Aetiological factors

Repeated pregnancies, food fads and fallacies, frequent infections, improper sanitary practices, poverty , unclean drinking water, poor dietary knowledge

Key Characteristics of Marasmus:

Severe Wasting:

Individuals with marasmus appear extremely thin, with a noticeable loss of muscle and subcutaneous fat. **Stunted Growth**:Children with marasmus experience significant delays in physical growth and development. **Weakened Immune System:**Marasmus compromises the body's ability to fight off infections, making individuals more susceptible to illnesses. **Developmental Delays:**The lack of essential nutrients can also lead to cognitive and behavioral developmental delays in children.

Other Symptoms:

These may include dry, brittle hair; sunken eyes; loose, wrinkled skin; and lethargy. **Weight Loss:**

Marasmus is characterized by significant weight loss, often below 60% of the expected weight for age.

Causes:

Marasmus can be caused by factors like inadequate feeding, physical defects preventing proper eating, diseases interfering with nutrient absorption, or infections causing loss of appetite.

Marasmic kwashiorkor is a severe form of malnutrition that combines features of both marasmus and kwashiorkor, resulting in a combination of wasting and edema. It's characterized by severe weight loss, muscle wasting, and fluid buildup in the body, particularly in the abdomen and extremities.

12. Explain why pellagra is known 3D disease. **[U] / [CO1, CO3)**

(2.5 mks ,2.5mks)

Pellagra is called the "3D disease" because it is characterized by three primary symptoms: dermatitis, dementia, and diarrhea. These symptoms arise from a deficiency of niacin (vitamin B3) in the diet.

Diarrhea (gastrointestinal mucositis)

Diarrhea occurs when the mucous lining of your intestines fails to replenish itself fast enough. Without this mucosa, the intestines can't digest properly, and they can't protect themselves from irritation and inflammation. Chronic inflammation affects the mucous lining along your entire gastrointestinal tract, from your mouth to bowels. This leads to gastrointestinal symptoms such as stomach pain and indigestion, as well as symptoms of oral mucositis, which include mouth sores and a red, swollen tongue.

Dermatitis (skin photosensitivity)

Pellagra also produces a distinctive type of dermatitis, particularly on sun-exposed areas of the body, such as the face, neck, arms, legs, hands and feet. Deprived of energy resources, the skin cells become more photosensitive. Dermatitis begins with rashes that resemble a sunburn, then progresses to rough, scaly, hyper-pigmented plaques. One telltale feature is a dark, hyper-pigmented collar around the neck. It's called Casal's collar, after the Spanish physician Don Gaspar Casal who first identified pellagra among the Spanish peasants in 1735.

Dementia (brain and nerve damage)

Pellagra is a systemic disease, affecting the body as a whole as cells are deprived of the energy they need to perform bodily functions. This will eventually affect the brain and nervous system. Early symptoms may be vague. They may include lethargy, apathy, and difficulty concentrating, or they may look more like anxiety or depression. Over time, symptoms progress to confusion and delusions. Severe cases may lead to permanent dementia, as well as neurological damage resulting in balance and coordination problems, muscle twitches and tremors**.**

13. List the digital tools used in nutrition therapy. **[K] / [CO1, CO4]**

(2.5mks, 2.5mks)

Definition of digital tools mark Importance of digital tools in nutrition therapy.

Digital tools play a crucial role in modern nutrition therapy, offering various applications for assessment, monitoring, and intervention. These tools include mobile apps, nutrient databases, and software for meal planning, all designed to enhance the efficiency and accuracy of nutritional guidance**.**

Digital tools used in nutrition therapy

1. Nutrition Tracking Apps:

Purpose: To monitor food intake, track calorie and nutrient consumption, and receive personalized recommendations.

Features: Food logging, database of nutrients, personalized recommendations, and goal setting.

2. Nutrient Databases:

Purpose: To assess the nutrient composition of foods and meals, aiding in meal planning and dietary analysis.

Examples: USDA Food Composition Database, FoodData Central.

3. Meal Planning Software:

Purpose:

To facilitate meal planning based on individual needs and preferences, including nutritional requirements and dietary restrictions.

4. Technology-Assisted Dietary Assessment:

Purpose: To improve the accuracy of dietary assessment through tools like mobile apps, cameras, and web-based platforms.

Examples: Interactive computer-based technologies, Personal Digital Assistants (PDAs), web-based technologies, mobile devices, and specialized cameras.

Benefits: Reduced errors, interviewer bias, and immediate data analysis.

5. Behavior Change Communication Tools:

Purpose: To provide support and encouragement to patients during their nutrition journey.

Examples: Online forums, chat groups, and motivational apps

Mobile Applications and AI-Based Nutrition Counselling

Smartphone applications assist users in tracking their daily food intake, providing instant feedback on calorie consumption, macronutrient distribution, and hydration levels.

AI-powered nutritionists analyse eating patterns and recommend diet adjustments based on individual needs and health goals.

Wearable Devices for Real-Time Monitoring

Devices such as smart watches and fitness trackers monitor physical activity, heart rate, and metabolic health indicators, complementing dietary recommendations.

Some devices integrate glucose monitoring for individuals with diabetes, helping them make real-time dietary adjustments.

Telehealth and Virtual Nutrition Counselling

Digital platforms enable remote consultations with registered dietitians and nutritionists, improving accessibility to professional guidance.

Virtual coaching allows for real-time support, motivation, and adjustments to diet plans.

Nutrigenomics and Personalized Diet Plans

Genetic testing combined with AI algorithms can tailor dietary recommendations based on an individual's genetic predisposition to certain health conditions.

This personalized approach enhances the effectiveness of nutrition**(** :

14. Explain on the methods of measurement of BMR. [U] / [CO1, CO2]

(2.5mks, 2.5mks)

Definition of BMR

Basal metabolic rate (BMR) can be measured using indirect calorimetry, which involves analyzing the amount of oxygen consumed and carbon dioxide produced by the body. Alternatively, BMR can be estimated using formulas that incorporate factors like age, sex, height, and weight.

Methods for Measuring BMR:

1. Indirect Calorimetry:

This method measures the oxygen consumption and carbon dioxide production of the body, which are indicators of energy expenditure. This technique requires specialized equipment and a controlled environment, making it more accurate than estimations.

2. Equations:

Several equations, such as the Mifflin-St. Jeor equation and the Harris-Benedict equation, can estimate BMR using age, sex, height, and weight. These equations provide a rough estimate but are less precise than direct measurement.

3. Direct Calorimetry:

This method involves measuring the amount of heat produced by the body in a sealed chamber. While highly accurate, it is less practical due to the equipment and time required.(bomb calorimeter)

15. Describe how the deficiency of vitamin A and vitamin D affects the body. [K] / [CO1]

(5mks)

Vitamin A deficiency happens when your body doesn't get enough vitamin A. Diet lacking in vitamin A and certain disorders can cause vitamin A deficiency. Symptoms include vision issues such as night blindness. Treatment involves taking vitamin A supplements.

Vitamin A deficiency can lead to many complications, including:

Eye problems: Vitamin A is a key factor in many functions of your eyes. Vision loss and blindness can occur. One of the early signs of vitamin A deficiency is night blindness (nyctalopia). If you have night blindness, you have trouble seeing well in the dark, but you can see normally if enough light is present. A retinal disorder causes night blindness.

As the vitamin A deficiency worsens, the whites of your eyes and your corneas can become dry and you aren't able to produce tears (xerophthalmia). Foamy spots called Bitot spots may appear in the whites of your eyes. Open sores on your corneas (corneal ulcers) may appear, or drying or clouding of your corneas (keratomalacia), which can lead to blindness.

Skin issues: A lack of vitamin A can cause dry, scaly and/or itchy skin.

Infertility: Vitamin A plays an important role in your reproductive system. Vitamin A deficiency can lead to trouble conceiving and infertility.

Growth issues: Vitamin A deficiency can cause delayed growth and development in children.

Respiratory tract infections: Vitamin A deficiency can cause problems with your immune system. This can lead to infections in your chest and throat.

Vitamin D

Vitamin D is an essential vitamin that your body uses for normal bone development and maintenance. Vitamin D also plays a role in your nervous system, musculoskeletal system and immune system.

Severe lack of vitamin D in children causes rickets. Symptoms of rickets include:

Incorrect growth patterns due to bowed or bent bones.,muscleweakness,bone pain, deformities in joints.

This is very rare. Children with a mild vitamin deficiency may just have weak, sore and/or painful muscles.

Lack of vitamin D isn't quite as obvious in adults. Signs and symptoms might include:

Fatigue, bone pain.,muscle weakness, muscle aches or muscle cramps.,mood changes, like depression.

In general, the two main causes of vitamin D deficiency are

Not getting enough vitamin D in your diet and/or through sunlight.

Your body isn't properly absorbing or using vitamin D.

16. Discuss the role of macronutrients (proteins, carbohydrates, and fats) in space nutrition

[U] / [CO1, CO4]

(2.5 mks, 2.5mks)

Space nutrition, also known as astronaut nutrition, refers to the specialized dietary needs and considerations for astronauts during space missions. It focuses on providing astronauts with adequate and balanced nutrition to support their health, well-being, and performance in the challenging space environment

Role of macronutrients in space nutrition

In space nutrition, macronutrients (proteins, carbohydrates, and fats) play crucial roles in maintaining astronaut health and performance. Proteins are vital for building and repairing tissues, while carbohydrates provide the primary source of energy. Fats serve as a concentrated energy source and also support various bodily functions**.**

Proteins:

In space, astronauts need adequate protein to support muscle maintenance and repair, which can be affected by the microgravity environment. Protein also helps in the production of hormones and chemicals

Carbohydrates:

Carbohydrates are the body's primary fuel source, especially for brain function and physical activity. In space, they are essential for maintaining energy levels during long-duration missions.

Fats:

Fats are a concentrated source of energy and also play a role in various physiological functions like hormone production and cell membrane structure. In space, they are important for energy storage and supporting overall health.

A balanced intake of macronutrients is essential for astronaut health in space for several reasons:

Energy Production:

All macronutrients contribute to energy production, but carbohydrates are the primary source, while fats provide a concentrated energy reserve.

Tissue Repair and Maintenance:

Proteins are crucial for building and repairing tissues, which is important in space where muscle and bone loss can occur due to microgravity.

Hormone and Chemical Production:

Proteins are involved in the production of various hormones and chemicals, which are necessary for regulating bodily functions in space.

Maintaining Overall Health:

A balanced diet ensures astronauts receive the necessary nutrients to maintain their health and well-being during long-duration space missions

Part C

Essay

Answer any 2 questions. Each question carries 10 marks

17. Prepare a note on vitamin D, Vitamin E and K [K] / [CO1)

(10mks)

Fat-soluble vitamins are vitamins A, D, E, and K. They are present in foods containing fats. The body absorbs these vitamins as it does dietary fats. They do not dissolve in water.

Vitamin D

Vitamin D is obtained through

naturally through exposure to sunlight or in the diet

through fortified foods

as supplements

The body obtains the compounds it needs to make vitamin D from food. It also produces vitamin D when ultraviolet (UV) light meets the skin.

Types

Vitamin D is not a single substance but a group of compounds collectively known as calciferol

The body absorbs calciferol into the bloodstream and then converts it to calcitriol.

Two types occur naturally:

vitamin D-3, found in animal fats

vitamin D-2, found in plants, such as mushrooms

Function

Vitamin D has two main roles in the body:

It maintains bone health.

It supports the immune system.

Dietary sources

A person can obtain some vitamin D from the sun, but most people will also need to use other sources, too. The main alternative is food.

Dietary sources include:

oily fish and fish oils,fortified dairy products, plant-based milks, and cereals beef liver eggs

Recommended intake

Current guidelinesTrusted Source recommend that people of all ages intake of 600 IU of vitamin D daily.

Deficiency

A vitamin D deficiency can affect:

older adults and children who do not spend much time out of doors

people with darker skin

some people with chronic health conditions

those who live far from the Equator, where winter days are short

those with obesity

The main effects of vitamin D deficiency include:

osteoporosis, or loss of bone mass

osteomalacia, when bones become soft

rickets, when a child's bones do not develop as they should

increased risk of infection and autoimmunity

Learn more about vitamin D deficiency.

Overdose

This can lead to:

nausea

headaches

low appetite and weight loss

a buildup of calcium in tissues and blood vessels

heart or kidney damage

high blood pressure

Vitamin E

Vitamin E is an antioxidant that can help the body destroy free radicals. Free radicals are unstable atoms that can cause oxidative stress. Oxidative stress can lead to cell damage, and this can result in cancer and other diseases. Vitamin E may help protect the body from a range of health issues.

Function

Some reasons why the body needs vitamin E are:

as an antioxidant

to boost the immune system

to dilate blood vessels and help prevent clotting

Dietary sources

Good sources of vitamin E include:

wheat germ oil,sunflower seeds and oil,almonds, hazelnuts, and peanuts,spinach and broccoli

kiwi fruit and mango

Deficiency

Vitamin E deficiency is rare, but it can affect people with Crohn's disease or cystic fibrosis. These conditions affect the liver's ability to absorb vitamin E.

A deficiency can result in:

nerve and muscle damage that affects movement and coordination

vision problems

a weakened immune system

As vitamin E is an antioxidant, a long-term deficiency could increase the overall risk of various diseases.

Vitamin K

Vitamin K helps the body form blood clots. Blood clotting is essential to prevent excessive bleeding.

Types

There are several types of vitamin K.

The two most common groups are:

Vitamin K-1 (phylloquinone), present in green, leafy vegetables and some other plant sources

Vitamin K-2 (menaquinones), present in animal sources and fermented foods

There are also synthetic forms and other forms that the body makes.

Function

Apart from blood clotting, vitamin K may also:

lower the risk of heart disease

enhance bone health

reduce the buildup of calcium in the blood

Learn more here about why we need vitamin K and where to get it.

Dietary sources

Food sources of vitamin K-1 and K-2 include:

kale

liver

spinach

parsley

butter

egg yolk

Deficiency

The body is not able to store as much vitamin K as it does vitamin A or D. This means a person needs a regular intake of vitamin K, and there is a higher chance of a deficiency.

A vitamin K deficiency may result in:

1.excess bleeding

2.lower bone density, in the long term

18.Write an essay on the signs and symptoms of protein deficiency. Elaborate on Prevention and Management of protein deficiency disorders [U] / [CO1]

(10mks)

Protein deficiency is when the intake is unable to meet body's requirements.Kwashiorkor and marasmus are the two main types of severe protein-energy undernutrition recognized by healthcare providers worldwide. The main difference between them is that kwashiorkor is predominantly a protein deficiency, while marasmus is a deficiency of all macronutrients — protein, carbohydrates and fats.

Kwashiorkor is a type of malnutrition characterized by severe protein deficiency. It causes fluid retention and a swollen, distended abdomen. Kwashiorkor most commonly affects children, particularly in developing countries with high levels of poverty and food insecurity. People with kwashiorkor may have food to eat, but not enough protein.

Kwashiorkor is one of the two main types of severe protein-energy undernutrition. People with kwashiorkor are especially deficient in protein, as well as some key micronutrients. Severe

protein deficiency causes fluid retention in the tissues (edema), which distinguishes kwashiorkor from other forms of malnutrition. People with kwashiorkor may look emaciated in their limbs but swollen in their hands and feet, face and belly. The distended abdomen typical of kwashiorkor can be misleading in people who are actually critically malnourished.

Kwashiorkor is rare in developed countries. It's mostly found in developing countries with high rates of poverty and food scarcity. Poor sanitary conditions and a high prevalence of infectious diseases also help set the stage for malnutrition. Kwashiorkor can affect all ages, but it's most common in children, especially between the ages of 3 to 5. This is an age when many children have recently transitioned from breastfeeding to a less adequate diet — one higher in carbohydrates but lower in protein and other nutrients.

The difference between kwashiorkor and marasmus

Kwashiorkor and marasmus are the two main types of severe protein-energy undernutrition recognized by healthcare providers worldwide. The main difference between them is that kwashiorkor is predominantly a protein deficiency, while marasmus is a deficiency of all macronutrients — protein, carbohydrates and fats. People with marasmus are deprived of calories in general, either because they're eating too little or expending too many, or both. People with kwashiorkor may not be deprived of calories in general but are deprived of protein-rich foods.

Symptoms and Causes

Edema (swelling with fluid, especially in the ankles and feet).

Bloated stomach with ascites (a build-up of fluid in the abdominal cavity).

Dry, brittle hair, hair loss and loss of pigment in hair.

Dermatitis — dry, peeling skin, scaly patches or red patches.

Enlarged liver, a symptom of fatty liver disease.

Depleted muscle mass but retained subcutaneous fat (under the skin).

Dehydration.

Loss of appetite (anorexia).

Irritability and fatigue.

Stunted growth in children.

Treatment

Marasmus

Marasmus is a severe form of malnutrition characterized by a deficiency of all macronutrients (proteins, carbohydrates, and fats), leading to a significant loss of body weight and tissue, particularly muscle and fat. It's primarily caused by a lack of sufficient calories and protein intake, resulting in energy deficiency.

Causes:

Inadequate calorie and protein intake: This can result from poverty, food shortages, or inadequate breastfeeding.

Poor food choices: Limited access to diverse and nutritious foods can contribute to marasmus.

Increased energy expenditure: Conditions like chronic infections, burns, or other illnesses can increase the body's energy needs, leading to marasmus if intake is not adequate.

Early weaning: Early weaning can result in inadequate feeding and starvation in infants.

Symptoms:

Extreme weight loss: Body weight is significantly reduced, often below 62% of normal weight for age.

Wasting: Muscle and fat tissue are depleted, leading to a thin, emaciated appearance.

Growth retardation: Children with marasmus may be stunted in size and development.

Irritability and lethargy: Lack of energy and reduced activity levels can be observed.

Changes in skin and hair: Skin may become dry and wrinkled, and hair can become thin and brittle.

Other signs: Bradycardia (slow heart rate), hypotension (low blood pressure), and hypothermia (low body temperature) can be seen.

Complications:

Slow growth and development: Marasmus can lead to long-term growth and developmental delays.

Increased susceptibility to infections: The weakened immune system makes individuals more vulnerable to infections.

Serious complications: Diarrhea, measles, and respiratory infections can be life-threatening in children with marasmus.

Treatment:

Rehydration and electrolyte balance:

Addressing dehydration and electrolyte imbalances is crucial.

Gradual refeeding:

Providing a balanced diet, starting with small amounts and gradually increasing the volume and nutritional content.

Addressing infections:

Treating any underlying infections is essential to improve overall health and recovery.

Monitoring:

Regular monitoring of weight, height, and other vital signs is needed to track progress and adjust treatment as necessary.

Key differences from Kwashiorkor:

Marasmus is a deficiency of all macronutrients (proteins, carbohydrates, and fats), while kwashiorkor is primarily a protein deficiency.

Kwashiorkor often presents with edema (swelling) due to low protein levels, while marasmus is characterized by muscle and fat wasting.

Marasmus is a serious condition that can be life-threatening if not treated promptly. Early diagnosis and intervention are essential to prevent complications and improve outcomes.

Marasmic Kwashiorkor

Marasmic kwashiorkor is a severe form of malnutrition that combines features of both marasmus and kwashiorkor. It occurs when a child with kwashiorkor (edematous malnutrition due to protein deficiency) also experiences a significant calorie deficiency. The result is a condition with both edema (swelling) and wasting (loss of muscle and fat mass).

Marasmic Kwashiorkor:

A child with marasmus who also develops edema due to protein deficiency is classified as marasmic kwashiorkor. This condition combines the wasting of marasmus with the edema of kwashiorkor.

Causes:

Marasmic kwashiorkor can develop when a child with kwashiorkor does not consume enough calories, or when a child with marasmus experiences a sudden increase in protein deficiency.

Significance:

Marasmic kwashiorkor is a serious condition with a high risk of complications and mortality, and requires immediate medical attention.

19. List the common types of vegetarian diets mentioned in the text. [K] / [CO1, CO4]

(5mks, 5mks)

A vegetarian diet contains no animal proteins (unless lacto-vegetarian or lacto-ovo vegetarian). A semi-vegetarian diet is a meal plan that contains little animal protein, but mostly plant-based

The main types of vegetarian diets include lacto-ovo vegetarian, lacto-vegetarian, ovo-vegetarian, and vegan. These differ based on the inclusion or exclusion of animal products like dairy and eggs.

Lacto-ovo vegetarian: This is the most common type, including plant-based foods, dairy, and eggs.

Lacto-vegetarian: This diet includes dairy products but excludes eggs and all meat.

Ovo-vegetarian: This diet includes eggs but excludes dairy products and all meat.

Vegan: This is the most restrictive type, excluding all animal products, including meat, poultry, seafood, eggs, and dairy.

Other common variations:

Flexitarian: This diet focuses on a mostly vegetarian diet, with occasional meat and fish consumption.

Pescatarian: This diet includes fish and other seafood, but excludes meat, poultry, dairy, and eggs.

Raw vegan: This diet focuses on raw, unprocessed plant-based foods.

Fruitarian: This diet primarily consists of fruits, nuts, and seeds

20. Explain about the mechanism of water balance.[U] / [CO1, CO2]

(5mks, 5mks)

Water balance in nutrition refers to the equilibrium between water intake and water output in the body. It's a crucial aspect of maintaining overall health and preventing dehydration or overhydration. Water intake comes from fluids and the water content in food, while water output occurs through urine, sweat, breathing, and faeces**.**

The body maintains water balance through a complex interplay of intake, output, and regulatory mechanisms like ADH and the renin-angiotensin-aldosterone system (RAAS). These mechanisms ensure that water levels in the body remain within a narrow, healthy range.

Fluid Intake:

Thirst is a primary drive for fluid intake, stimulated by dehydration or increased sodium concentration in the blood. The hypothalamus, particularly the osmoreceptors, plays a crucial role in detecting these changes and triggering the thirst sensation.

Water Output:

The kidneys are the primary organs for water excretion, adjusting urine concentration to reflect the body's water needs. Antidiuretic hormone (ADH), also known as vasopressin, increases water reabsorption in the kidneys, conserving water when the body is dehydrated.

Renin-Angiotensin-Aldosterone System (RAAS):

When blood volume decreases, the kidneys release renin, which activates the RAAS. This system ultimately increases blood pressure and sodium reabsorption in the kidneys, leading to increased water retention.

Kidney Function:

The kidneys filter blood and produce urine, a process that helps regulate water and electrolyte balance. They can adjust the amount of water reabsorbed back into the bloodstream based on the body's needs.

1. Dehydration:

When the body loses water (e.g., through sweating, urination, or vomiting), blood volume decreases, and the osmolality of the blood increases.

2. Thirst and ADH Release:

Osmoreceptors in the hypothalamus detect the increased osmolality and trigger the thirst sensation, leading to increased fluid intake. Simultaneously, ADH is released from the pituitary gland, causing the kidneys to reabsorb more water.

3. RAAS Activation:

Decreased blood volume also triggers the release of renin from the kidneys, initiating the RAAS.

4. Sodium and Water Reabsorption:

The RAAS ultimately leads to increased sodium reabsorption in the kidneys, and water follows the sodium back into the bloodstream.

5. Water Output Adjustment:

The kidneys adjust urine concentration based on the body's needs, producing more concentrated urine when dehydrated and more dilute urine when the body has excess water.